

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Echocardiographic variables were abstracted from TTE reports. WebOMR was used for medical chart review. Study data were collected and managed using Castor Electronic Data Capture.
Data analysis	An AI HFpEF model (EchoGo Heart Failure, Ultromics Ltd) was used to analyze the data in this study. All analyses were performed using R version 4.2.3 (R Foundation, Vienna, Austria).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data supporting this study contain protected health information and thus are not provided in a public repository but are available from the corresponding author upon reasonable request. BIDMC's institutional TTE database (Viewpoint 6 Version 6.12) was used to identify cases and controls for the study

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Sex was considered in the study design. The findings of this study apply to both sexes. The sex variable is defined as self-designated at the time of echocardiogram as male or female. Consent for the sharing of individual-level data was waived by the Institutional Review Board at BIDMC. In table 1 of the main text, 141 female sex controls [256] and 130 female sex cases [240] were listed with a p-value of 0.910 under demographics. In table 8 of the Supplemental Appendix, AI HFpEF model lists under female subgroup: 35 intermediate, 86% sensitivity, 53.3% specificity, 80.2% NPV, 63.2% PPV. Under the male subgroup: 40 intermediate, 87.4% sensitivity, 65.6% specificity, 83.1% NPV, and 72.8% PPV. The H2FPEF model lists under female subgroup: 165 intermediate, 95.6% sensitivity, 36.8% specificity, 82.4% NPV, and 73% PPV. Under the male subgroup, 141 intermediate, 100% sensitivity, 43.8% specificity, 100% NPV, and 74.3% PPV. The HFA-PEFF model lists under female subgroup: 153 intermediate, 100% sensitivity, 72.1% specificity, 100% NPV, and 86.2% PPV. For the male subgroup: 116 intermediate, 98.5% sensitivity, 77.3% specificity, 96.1% NPV, and 86.5% PPV.

Reporting on race, ethnicity, or other socially relevant groupings

Race was defined as a self-designated value listed in medical record (White, Black, Asian, or Other) and was used to determine the model classification performance.

Population characteristics

Among cases, the mean $\pm$ SD age was 74.2 ± 12.1 years and 54.2% were female. Among controls, the mean $\pm$ SD age was 75.0 ± 13.0 years and 55.1% were female. Clinical covariates included age, sex, Body mass index, Race, Hypertension, Hyperlipidemia, Atrial fibrillation, Diabetes mellitus, Chronic kidney disease, Coronary artery disease, Chronic obstructive pulmonary disease, NYHA class, Angina, Smoking, Anemia, History of stroke/TIA, History of MI, History of CABG, Presence of pacemaker.

Recruitment

Individuals who received a TTE between 2018 and 2022 at BIDMC, a large tertiary care hospital of Harvard Medical School in Boston, Massachusetts, were considered for inclusion in a retrospective case-control study. This time window was chosen to reflect contemporary management of HFpEF and permit abstraction of structured TTE data on biplane left ventricular ejection fraction (LVEF), not available prior to 2018. Only an individual's first TTE during the study window was considered. All patients were required to have a biplane LVEF 50% on the index TTE and have technically evaluable images. The focus on biplane LVEF was for several reasons: 1) there may be possible differences across LVEF techniques, 2) current American Society of Echocardiography guidelines advocate for use of biplane LVEF, and 3) the criteria for ejection fraction in the development of the AI HFpEF model used biplane LVEF values.

Ethics oversight

The study protocol was approved by the Institutional Review Board at BIDMC.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

An a priori sample size estimation was completed consistent with the original validation of the AI model (Akerman et al., JACC Adv, 2023). The performance benchmarks generated for the original validation of the model (74% and 65% for sensitivity and specificity, respectively), were used as a minimum acceptable performance. These were tested using a One-sided Binomial Exact Test. The observed sensitivity and specificity from the original validation trial (Akerman et al., JACC Adv, 2023) were used to estimate likely performance in a separate external validation cohort; sensitivity (87.8%; 95% CI: 84.5%-90.9%) and specificity (81.9%; 95% CI: 78.2%-85.6%). We assumed that the lower bound of the confidence interval was an accurate representation of likely performance in a different cohort (i.e., 85% and 78%, respectively), and the lower bound of an anticipated confidence interval approximately 10% lower. Thus, our anticipated performance for sensitivity (lower bound of confidence interval) was 75%, and for specificity was 68%.

We therefore completed a sequence of simulations of One-sided Binomial Exact Tests assuming a sample size range of 100 to 400, an observed sensitivity of 75% to 90%, and observed specificity of 68% to 85%. The upper bound of these simulations was generated from the upper bound of the confidence interval from the previous validation. A full grid of permutations of these values for sensitivity and specificity was generated, resulting in 8,928 possible permutations of sensitivity and specificity performance. For each permutation, two One-sided Binomial Exact Tests compared the relevant sensitivity to the lowest acceptable benchmark (74%), and likewise for specificity (65%). We then compared simulations of significant One-sided Binomial Exact Tests with the feasibility of obtaining cases from the independent site (Beth Israel Deaconess Medical Center, Boston, Massachusetts, USA), and decided that approximately 300 cases and matched 300 controls would permit a maximum of 20% dropout due to uncertain outputs, and still significantly ($p < 0.025$) higher performance than the benchmark for a wide range of observed sensitivity (80%-90%) and specificity (71.2%-85%) permutations. A final sample of 240 cases and 256 controls were identified after application of exclusions.

Data exclusions	In individuals that received TTEs at BIDMC from 2018-2022, those with missing biplane LVEF (N = 26,735) or biplane LVEF <50% (N = 3,631) were excluded. In the eligible controls (no presence of HF), one individual (N = 1) that could not be exactly matched to cases on age and sex was excluded. Out of the controls that were matched, those with poor technical image quality (N = 43) were excluded. Among the cases (individuals with grade II or III diastolic dysfunction and presence of HF), individuals that were not randomly sampled (N = 323) were excluded from the study. One individual (N = 1) that could not be exactly matched after random sampling was also excluded. Cases that were exactly matched to controls on age and sex but had poor technical image quality (N = 59) were further excluded from the study.
Replication	The findings represent an external validation of a previously derived AI HFpEF model. In several iterations of running this model on the same images, results were identical.
Randomization	Selection of cases and controls was non-random. Cases were identified through query of the institutional TTE database as having grade II-III diastolic dysfunction. Additionally, cases were required to have evidence of recent hospital admission for heart failure (HF) within a year prior to the index TTE date, as evidenced by a primary discharge diagnosis of 428.X (under the International Classification of Diseases, 9th Revision [ICD-9] classification) prior to October 1, 2015 or I50.X (under the International Classification of Diseases, 10th Revision [ICD-10] classification) on or after October 1, 2015. Controls were similarly identified as having grade I, or no diastolic dysfunction on TTE, and no evidence of a hospital admission for HF within one-year prior to and after the index TTE date. Presence or absence of HF hospitalization was additionally adjudicated and confirmed at the time of manual chart review, but this information was not used in case or control selection. Among possible cases and controls, 300 cases were randomly selected and exactly matched on age, sex, and year of index TTE to control subjects. No covariate adjustment was performed as the intention was to identify how the AI HFpEF model predicted outcomes agnostic to these data.
Blinding	Patient status (case or control) was not blinded to analysts during extraction of relevant clinical parameters and patient outcomes. However, the AI HFpEF model was agnostic to case or control status.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	This study is not a clinical trial.
Study protocol	This study is not a clinical trial.
Data collection	Data collection took place within BIDMC. Data collection took place between 2018 and 2022, with recruitment occurring from June 2022 to August 2022.
Outcomes	The outcome measures for this study included all-cause mortality and HF hospitalization within the year following the index TTE, with occurrence of an outcome abstracted by medical chart review.

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.